

Q1. GlobalTech Ltd. has its head office in London and a regional office in Singapore. The Singapore server synchronizes its clock with the London NTP server. During one synchronization exchange, the timestamps below are recorded in milliseconds according to each machine's own clock:

Timestamp	Meaning
$t_1 = 1000.0$ ms	Singapore client sends the NTP request
$t_2 = 1030.0$ ms	London server receives the request
$t_3 = 1030.0$ ms	London server sends the reply
$t_4 = 1020.0$ ms	Singapore client receives the reply

Assuming symmetric network delay and standard NTP notation, compute the round-trip delay and the clock offset of the Singapore client relative to the London server.

Q2. A healthcare data analytics organization is migrating its patient records, medical imaging metadata, and sensor-generated monitoring logs to a Hadoop-based storage and processing cluster.

The organization currently has 6,200 TB of raw data that must be stored in HDFS. The data team expects an additional 300 TB of new data to be generated and ingested during the next six months.

After ingestion, several MapReduce and Spark jobs will be run for data cleansing, transformation, and reporting. These jobs are expected to generate output datasets equal to 20% of the available data.

For storage capacity planning, assume the following:

- Intermediate data factor = 0.30
- Compression rate = 2.5
- HDFS replication factor = 3

I. Calculate the total logical data size that must be stored after including the expected six-month data growth.

Ans. Given

II. Determine the effective Hadoop storage requirement after accounting for:

- MapReduce/Spark output data
- Intermediate processing data
- Compression
- HDFS replication

III. Calculate the storage expansion factor, i.e., how many times larger the final Hadoop storage requirement is compared to the total logical data size calculated in Part I.

IV. If each DataNode contributes 120 TB of usable disk capacity, estimate the minimum number of DataNodes required to store the data safely.

Q3. A multinational company, DataSphere Inc., has its headquarters in New York and a branch office in Tokyo. The Tokyo server synchronizes its clock with the New York NTP server.

During one NTP synchronization exchange, the following timestamps are recorded (in milliseconds according to each machine's own clock):

Timestamp	Meaning
$t_1 = 2500.0$ ms	Tokyo client sends the NTP request
$t_2 = 2545.0$ ms	New York server receives the request
$t_3 = 2550.0$ ms	New York server sends the reply
$t_4 = 2520.0$ ms	Tokyo client receives the reply

Assuming symmetric network delay and standard NTP notation:

1. Calculate the round-trip delay (δ).
2. Calculate the clock offset (θ) of the Tokyo client relative to the New York server.
3. Determine whether the Tokyo client's clock is ahead of or behind the New York server's clock, and by how many milliseconds.

Ans.

Q4. A distributed video analytics system runs on 32 worker nodes using a task-based scheduler. Each video frame passes through the following three stages:

Stage	Fraction of total work	Parallelism characteristics
Feature Extraction	0.45	perfectly parallelizable
Object Tracking	0.35	20% serial (coordination), 80% parallelizable
Result Aggregation	0.20	fully serial (global ordering)

Assume:

- The workload is perfectly balanced across all 32 nodes (no worker remains idle).
- Communication overhead is negligible.
- The workload scales linearly with problem size, so increasing the number of frames keeps all 32 nodes busy.

Questions

1. Using Amdahl's Law, calculate the maximum overall speedup achievable on 32 nodes.
2. Determine the parallel fraction of the entire application.
3. Using Gustafson's Law, compute the projected speedup on 32 nodes when the workload scales with problem size.
4. Compare the results from Amdahl's Law and Gustafson's Law and briefly explain why they differ.